

Question C.2 — A Deterministic Certificate via the Peak Statistic

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1 The Two Units, and What a Direct Comparison Erases

Question addressed. What is the physical unit of τ_{ARF} , and that of the threshold λ ? How does comparing them directly sidestep the sequential nature of the detector?

The two units. In a stream, there is no physical clock: the only time variable available is the index of the incoming example, so τ_{ARF} is measured in *observations*. The threshold, meanwhile, is read off the recurrence of the cumulative detector¹

$$S_t = \max(0, S_{t-1} + e_t - p_0 - \delta_P), \quad (1)$$

where $e_t \in \{0, 1\}$ flags an error at step t , p_0 the normal pre-drift error rate, and δ_P a tolerance subtracted at every step. Since e_t is 0 or 1, an error pushes S up by nearly one unit while a success barely brings it down: up to the tolerance, S_t counts the errors accumulated beyond the baseline. Since λ only ever appears in the comparison $S_t \geq \lambda$, it must be read in the same unit — $\lambda = 50$ does not mean “wait 50 steps”, but “wait until roughly 50 excess errors have accumulated”. The two quantities are therefore counts of different natures: writing $\tau_{\text{ARF}} < \lambda$ makes no dimensional sense.

What a direct comparison hides. Comparing the two amounts to treating λ as a duration: “it takes λ errors, hence so many steps”. This shortcut assumes evidence accumulates at a fixed rate — without an error-per-step rate, there is no way to convert one unit into the other — and the assumption is baked into the very act of comparing, whether or not the conversion is written out. But no such rate exists: the detector receives errors one at a time and does not output a number, but a date — the one at which the trajectory of S_t crosses λ . Three features of that trajectory disappear once it is replaced by a constant-slope ramp.

The rate slows down. Right after the drift, the counter’s increment has expectation $\Delta e - \delta_P$; but the forest repairs itself, the excess error drifts back toward the baseline, and this gain dries up. The manuscript instead assumes a constant excess of Δe throughout the whole transient, then none at all — a rectangle, whose height is precisely the fixed rate the comparison requires.

Acquired evidence can be destroyed. When S_t would go below zero, the $\max(0, \cdot)$ in (1) resets it to 0: this is not a pause, everything accumulated so far is lost. The same errors, clustered right after the drift or spread out over the horizon, therefore do not yield the same alarm date, whereas an average rate makes no distinction between the two. This is the precise meaning of “sequential”: the date depends on the order of arrival, not only on the total.

The alarm may never come. Over the campaign’s horizon $H = 2000$, the threshold may never be crossed².

Conclusion. An error count and an observation count cannot be compared by positing a rate. Homogeneity is restored the other way around: by bringing both quantities into the unit of evidence, where the comparison is threshold against accumulated evidence, on each run and without averaging. That is the object of the peak statistic $S_{\max}(H)$ and the deterministic certificate that follows from it, developed in Section 2 below.

2 A Deterministic Certificate: The Peak Statistic

Question addressed. Show that the CUSUM’s peak statistic can be written $S_{\max}(H) = \max_{0 \leq k \leq j \leq H} [A(k, j) - (j - k) \delta_P]$, the maximum ranging over all sub-windows $[k, j] \subseteq [0, H]$.

Notation. Post-drift steps are indexed $t = 1, \dots, H$, with $t = 0$ the drift instant itself. Following the statement, set

$$A(k, j) = \sum_{t=k+1}^j (e_t - \hat{p}_0), \quad X_t = e_t - \hat{p}_0 - \delta_P, \quad S_0 = 0, \quad S_t = \max(0, S_{t-1} + X_t),$$

and $S_{\max}(H) = \max_{0 \leq t \leq H} S_t$. The baseline here is the *estimated* baseline $\hat{p}_0$, as throughout Question C. By convention an empty sum is 0, so $A(j, j) = 0$.

¹Equation (3) of the manuscript, section III-D (“The Starvation Effect”), notation kept as is. The R2 protocol creates the detector at the drift instant, hence $S_{\tau^*} = 0$; the manuscript’s Proposition 3 instead keeps $S_{\tau^*} = s_0 \geq 0$ — setting $s_0 = 0$ here is a specialization of the protocol.

²Beyond the horizon the crossing eventually happens, but its mean delay grows exponentially with λ (manuscript, Remark 4): the alarm then no longer bears on the drift.

Proposition 1 (Lindley’s closed form). *For every $j \in \{0, \dots, H\}$,*

$$S_j = \max_{0 \leq k \leq j} \sum_{t=k+1}^j X_t = \max_{0 \leq k \leq j} [A(k, j) - (j - k) \delta_P]. \quad (2)$$

Taking the maximum over j gives $S_{\max}(H) = \max_{0 \leq k \leq j \leq H} [A(k, j) - (j - k) \delta_P]$.

Proof. By induction on j .

Base case. For $j = 0$, the only admissible index is $k = 0$ and the corresponding sum is empty, hence zero: the right-hand side equals $0 = S_0$.

Inductive step. Assume (2) holds at rank $j - 1$. Then

$$S_j = \max(0, S_{j-1} + X_j) = \max\left(0, \max_{0 \leq k \leq j-1} \sum_{t=k+1}^{j-1} X_t + X_j\right) = \max\left(0, \max_{0 \leq k \leq j-1} \sum_{t=k+1}^j X_t\right),$$

the last equality following from X_j not depending on k , so it may enter the maximum. Now the term 0 is exactly the empty sum obtained for $k = j$: the two arguments of the outer maximum merge into a single maximum over $0 \leq k \leq j$, giving (2) at rank j .

Conclusion. The second form follows by expanding the sum:

$$\sum_{t=k+1}^j X_t = \sum_{t=k+1}^j (e_t - \hat{p}_0) - (j - k) \delta_P = A(k, j) - (j - k) \delta_P,$$

since δ_P is subtracted once per step and the sub-window $[k, j]$ contains $j - k$ of them. Finally $S_{\max}(H) = \max_{0 \leq j \leq H} S_j = \max_{0 \leq j \leq H} \max_{0 \leq k \leq j} [\cdot] = \max_{0 \leq k \leq j \leq H} [\cdot]$. $\square$

Corollary 1 (The certificate). *Fix $\lambda > 0$. Then*

$$S_{\max}(H) \geq \lambda \iff \exists 0 \leq k \leq j \leq H \text{ such that } A(k, j) - (j - k) \delta_P \geq \lambda.$$

In particular, the existence of a single sub-window meeting this bound guarantees that the alarm has fired by time H — a statement true on every individual run, with no averaging and no distributional assumption.

Proof. Immediate from Proposition 1: $S_{\max}(H)$ is, by definition, the maximum of $A(k, j) - (j - k) \delta_P$ over all admissible (k, j) , so this maximum is $\geq \lambda$ if and only if at least one of its terms is. $\square$

Necessary and sufficient, versus only sufficient. The equivalence in Corollary 1 holds only because it ranges over *every* sub-window. Restricting the check to the single full window $(k, j) = (0, H)$ gives the one-directional statement used below:

$$A(0, H) - H \delta_P \geq \lambda \implies S_{\max}(H) \geq \lambda,$$

which is only *sufficient*: it is one particular term in the maximum, so clearing the bar on the full window certainly clears it overall, but failing to clear it proves nothing — a burst of evidence concentrated in the middle of the horizon can be diluted, in the full-window average, by quiet periods before or after it, and still satisfy the two-sided condition on some shorter sub-window. This is exactly why anchoring the certificate on a short window near the drift (Section 3) can succeed where the full-window version fails, without any contradiction.

What the identity makes visible. The reset to zero is not a technical correction: it is *equivalent* to the freedom to choose the starting point. A CUSUM that restarts from zero whenever the counter would go below the line does nothing other than retain, at every instant, the best recent sub-window. This is why the maximum ranges over pairs (k, j) rather than over j alone: the detector does not measure the evidence accumulated since the drift, but the evidence accumulated over the most favorable sub-window.

Two immediate consequences. First, $S_{\max}(H) \geq 0$, since the choice $k = j$ is always available. Second, and more importantly, the identity is *deterministic*: it holds on each individual run, with no averaging or distributional assumption. This is what licenses Corollary 1, applied to the campaign in Section 3 below (and reused for the full-window discussion in Q C.3).

Numerical check. The closed form (2) was checked against the naive recurrence on the 2000 trajectories of the campaign: maximal deviation 1.26×10^{-11} , i.e. machine precision. The analysis script refuses to proceed further if this check fails.

3 Application to the Campaign

Both directions of Corollary 1 are now confronted with data: the one-directional full-window certificate first, then the short-window one, which the corollary makes necessary and sufficient for $S_{\max}(w) \geq \lambda$ and hence sufficient for detection within H .

Full window. With $H = 2000$ and $\delta_P = 0.01$, the full-window requirement $A(0, H) \geq \lambda + H\delta_P$ demands a toll of $H\delta_P = 20$ regardless of λ , i.e. raw areas of 28, 45 and 70 for $\lambda \in \{8, 25, 50\}$. Against the evidence budget of about 18 established in Q C.3, these are met only exceptionally: over the 2000 runs, $A(0, H)$ reaches 28 in 473 cases, 45 in 66, and 70 in none. The full-window certificate therefore proves detection for 23.7% of runs at $\lambda = 8$ (Wilson 95% interval [21.8; 25.6]), 3.3% at $\lambda = 25$ ([2.6; 4.2]) and none at $\lambda = 50$ ([0.0; 0.2]), and at the weakest amplitude ($\Delta e = 0.028$) for no run at all. Being only sufficient, this failure does not mean the alarm never fires; it only means the full window is the wrong place to look for the proof.

Short window. Anchoring instead on a window $w = \text{round}(3 \times 18.5/\Delta e)$ steps right after the drift — sized to the transient rather than to the whole horizon — and checking the *quantified* condition on that window, i.e. “some sub-window of $]0, w]$ satisfies the certificate inequality $A(k, j) - (j - k)\delta_P \geq \lambda$ of Corollary 1”, equivalently $S_{\max}(w) \geq \lambda$, gives a very different picture³.

Δe	w	$\lambda = 8$		$\lambda = 25$		$\lambda = 50$	
		certificate	detected	certificate	detected	certificate	
0.028	1 969	0.10 [0.06;0.17]	0.10 [0.06;0.17]	0.00 [0.00;0.04]	0.00 [0.00;0.04]	0.00 [0.00;0.04]	
0.085	650	0.95 [0.89;0.98]	0.96 [0.90;0.98]	0.09 [0.05;0.16]	0.15 [0.09;0.23]	0.00 [0.00;0.04]	
0.141	394	1.00 [0.96;1.00]	1.00 [0.96;1.00]	0.44 [0.35;0.54]	0.70 [0.60;0.78]	0.00 [0.00;0.04]	
0.194	287	1.00 [0.96;1.00]	1.00 [0.96;1.00]	0.73 [0.64;0.81]	0.91 [0.84;0.95]	0.00 [0.00;0.04]	
0.243	229	1.00 [0.96;1.00]	1.00 [0.96;1.00]	0.78 [0.69;0.85]	0.94 [0.88;0.97]	0.00 [0.00;0.04]	
0.361	154	1.00 [0.96;1.00]	1.00 [0.96;1.00]	0.57 [0.47;0.66]	0.83 [0.74;0.89]	0.00 [0.00;0.04]	
0.498	112	1.00 [0.96;1.00]	1.00 [0.96;1.00]	0.02 [0.01;0.07]	0.02 [0.01;0.07]	0.00 [0.00;0.04]	

Brackets: Wilson 95% interval, $n = 100$ seeds per amplitude.

Reading the table. Columns report, per amplitude, the fraction of the campaign’s 100 seeds where the short-window certificate fires versus the fraction where the alarm was actually detected within H (the $\lambda = 50$ detected column is omitted: it never exceeds 0.01, consistent with the near-total blind spot already reported for this threshold). Two things hold across every one of the 60 $(\Delta e, \lambda)$ cells computed, not only the seven shown: the certificate *never* exceeds observed detection — exactly what sufficiency requires — and for $\lambda = 8$ it very nearly *equals* it (mean gap 0.0005, median 0, over the 20 amplitudes; across the 60 cells of the three thresholds the mean gap is 0.037), meaning the mathematical guarantee is almost never more pessimistic than reality at this threshold. For $\lambda = 25$ the gap widens (e.g. 0.44 vs. 0.70 at $\Delta e = 0.141$), and for $\lambda = 50$ the short window essentially never supplies the required evidence at all — the window is sized to the transient, and 50 excess errors is more than a typical transient of this length can generically offer, even though the alarm can still fire later through paths this particular certificate does not capture.

³Restricting instead to the single sub-window $]0, w]$, i.e. $A(0, w) \geq \lambda + w\delta_P$, is a strictly stronger requirement and yields noticeably lower figures at $\lambda = 25$: 0.39 instead of 0.44 at $\Delta e = 0.141$, 0.30 instead of 0.40 at $\Delta e = 0.416$. The tabulated values are the quantified ones, as computed by `analyse_QCD.py`.